

Data Exploration and Correlation Analysis

Student Mental Health and Academic Performance Analysis Pipeline We Would Follow

We first compute the full **Pearson correlation matrix** among all 21 features (mental-health scores, academic grades, social support indicators, etc.) to identify co-occurring patterns. For example, we expect positive correlations among stress and depression measures and negative correlations between those and performance indicators. Prior work shows that higher symptom scores tend to align with lower grades bmcpublichealth.biomedcentral.com. In our pipeline we would visualize this with a heatmap or network graph. For instance, the figure below (a correlation-network from related student data) highlights tightly linked clusters of variables: depression, anxiety, exhaustion, cynicism and self-efficacy are all interrelated and jointly impact academic efficacy nature.com. In practice, we would place `stress_level`, `depression`, `academic_performance`, and other scale features into a matrix and plot it to spot strong associations. This preliminary analysis confirms, for example, a significant **negative correlation between poor mental health and academic achievement** bmcpublichealth.biomedcentral.com, guiding our choice of predictors for modeling.

Correlation network among student mental-health and academic variables (e.g. depression, anxiety, exhaustion, cynicism, efficacy). A clique of related factors (blue nodes) strongly co-vary and relate to performance nature.com.

Predictive Modeling of Stress, Depression and Performance

We then build separate supervised models for each target: `stress_level` (ordinal 0–2), `depression` (0–27 score), and `academic_performance` (0–5 grade scale). After splitting the data (e.g. 80/20 train/test) and appropriate preprocessing (handling missing values, scaling, encoding), we train a suite of algorithms (e.g. logistic/ordinal regression, decision trees, random forests, SVM, gradient boosting, neural networks) with cross-validation and hyperparameter tuning. Model performance is evaluated using appropriate metrics: for classification (stress level, perhaps binned depression) we use accuracy, precision, recall, F1, and ROC-AUC; for regression (depression score or performance points) we check R^2 , RMSE, etc. In prior student mental-health studies, cross-validated random forests and boosting often excel: for instance, one framework achieved $AUC > 0.90$ on depression prediction papers.ssrn.com. Similarly, we might find decision trees or forests give the best stress-level accuracy while SVM/boosting excel at predicting depression severity pdfs.semanticscholar.org.

- **Training and Evaluation:** We apply k-fold (e.g. 10-fold) cross-validation to assess generalization bmcpublichealth.biomedcentral.com. For each fold we compute metrics (accuracy, precision, recall, F1) on held-out data bmcpublichealth.biomedcentral.com. Early experiments might show, for example, `stress_level` classified with ~80% accuracy by a Random Forest, and `depression` scored with high correlation by a Gradient Boosting Machine (consistent with the literature’s reported $ROC \sim 0.90$ papers.ssrn.com).
- **Best Algorithms:** Drawing on [10], [12], [44], etc., we expect tree-based ensembles (RF, XGBoost) and SVM to perform strongly. For example, Mutalib et al. found a Decision Tree gave the best stress predictions and an SVM best for depression in student data pdfs.semanticscholar.org. These methods naturally provide feature importance and handle mixed data types.
- **Results Interpretation:** We would report each model’s accuracy or R^2 and confusion matrices. If needed, we might discretize the depression score into “none/mild/moderate/severe” levels (as is standard with PHQ-9) and similarly treat performance as ordered categories. All results are visualized (e.g. ROC curves, prediction vs actual plots) and summarized in text and tables.

Feature Importance and Interpretation

To understand **which factors drive each prediction**, we compute feature importance. For tree-based models we use e.g. mean decrease in Gini (Random Forest) or SHAP values; for linear models we use coefficient magnitudes. For instance, in predicting depression risk a large study identified suicidal ideation, anxiety level, sleep quality and academic stress as the top predictors bmcpublichealth.biomedcentral.com. We expect similar key features: likely high stress scores, low perceived social support, and poor sleep or health symptoms will rank highly. We present these as bar charts or importance plots. For example, a chart might show “Stress Score – 0.24”, “Sleep Quality – 0.15”, “Family Support – 0.12”, etc. These rankings inform domain experts which aspects (academic pressure, social environment, physical symptoms) most influence each outcome bmcpublichealth.biomedcentral.com. All code (e.g. scikit-learn’s `feature_importances_` or SHAP) and charts are documented for reproducibility.

Clustering: Student Profiles

We apply unsupervised **clustering** (e.g. k-means, hierarchical clustering) to the students based on all mental-health, performance and contextual features, aiming to identify natural “profiles.” To choose k, we use the elbow method or silhouette score on the distance matrix (features may be normalized or PCA-reduced). Typical solutions might yield 3–5 clusters. Each cluster is then profiled by its centroid or average feature values. For example, one cluster might have *low stress, high grades, strong social support*, while another shows *high stress, low performance, low support*. Prior research confirms this: Liu et al. found three clusters (low/mild/high anxiety risk) among ~10,000 students, with the high-risk group marked by heavier academic pressure and poorer grades bmcpsy psychiatry.biomedcentral.com. We would label our clusters similarly (e.g. “Low Stress/High Performance”, “High Stress/Low Support”). The clustering process and resultant profiles are visualized (e.g. cluster-means bar charts or feature heatmap), and validated by checking that clusters differ significantly on key outcomes. Such segmentation can help tailor interventions to each profile.

Association network from related student data (Fig.11 in [25]). It reveals how variables like stress, social vulnerability, gender, etc., form linked subgroups nature.com. In practice, we might similarly identify clusters (e.g. “low-risk”, “mild-risk”, “high-risk” profiles) with k-means bmcpsy psychiatry.biomedcentral.com.

Integration into a Monitoring System

Finally, we outline how to embed this pipeline into a software system for real-time student support. The multi-stage outputs (risk scores, feature importances, cluster membership) would feed a backend decision-support service. For example, a daily data ingestion job could update each student’s profile from surveys or activity logs; the ML models compute updated stress/depression risk and performance forecasts; and alerts/visualizations are pushed to an educator dashboard or counselor app. Architecturally, this resembles the **multi-layer predictive framework** proposed in recent literature: Selim (2025) describes a system integrating survey tools, behavioral analytics, and sensor data with privacy-preserving preprocessing and human-in-the-loop review papers.ssrn.com. We would similarly combine self-reports (questionnaires), academic records, and optional passive data, feeding supervised models and clustering in the cloud.

Key integration points include:

- **Alerting and Dashboards:** Whenever a student’s predicted risk exceeds a threshold (e.g. high stress + low support), the system flags this case in a dashboard. As noted in [20], real-time monitoring tools can give educators early warnings about students showing stress or anxiety changes nature.com. For instance, an automated message could notify a counselor that a particular student’s model now predicts “High Stress” with 85% probability.
- **Privacy and Oversight:** We incorporate best practices – e.g. encryption of personal data, consent flows, and a human review step before any intervention papers.ssrn.com. The models are decision-

support, not decision-makers. We also log model confidence and explanations (feature importance) so counselors understand *why* a student is flagged (e.g. “Academic pressure and low family support contributed” papers.ssrn.com bmcpublichealth.biomedcentral.com).

- **Continuous Learning:** As more data arrives, the models can be retrained periodically to adapt to changing student populations. We might also gather feedback (e.g. actual counseling outcomes) to refine the algorithms.

In summary, our pipeline outputs a correlation matrix visualization, three predictive models (with accuracy/F1 and importance charts), and a clustering of students into typologies. These components can be exposed through an API or integrated into a **student wellness platform**. For example, a teacher’s web interface could display each student’s current stress/depression risk scores and cluster label, along with the top contributing factors. This meets both research goals (by providing statistical analyses and validated models) and application needs (by enabling real-time monitoring and intervention) nature.com papers.ssrn.com.

Sources: We use established methods and cite recent studies for each step. Correlation analysis is standard (e.g. Pearson’s r in health research bmcpublichealth.biomedcentral.com bmcpublichealth.biomedcentral.com). Predictive modeling follows common

ML practice (tree ensembles, SVM, cross-validation bmcpublichealth.biomedcentral.com pdfs.semanticscholar.org). Feature importance and clustering approaches are validated in the literature bmcpublichealth.biomedcentral.com bmcpublishing.biomedcentral.com. Finally, system integration is informed by recent frameworks for digital mental-health monitoring nature.com papers.ssrn.com papers.ssrn.com. These citations support our methodological choices and the interpretation of results.